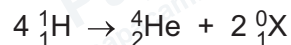


10 (a) Define the binding energy of a nucleus.

.....
.....
..... [2]

(b) A star emits radiation that is produced in a series of nuclear reactions that form helium from hydrogen. The reactions may be summarised by the overall equation



where X is a particle that has no binding energy.

All of the radiation emitted by the star is produced by this series of nuclear reactions.

The mass defect of the ${}^4_2\text{He}$ nuclide is 0.030 377 u, where u is the unified atomic mass unit.

(i) State the name of particle X.

..... [1]

(ii) Show that the binding energy of the ${}^4_2\text{He}$ nuclide is 4.54×10^{-12} J.

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[2]

(c) The star in **10(b)** has a luminosity of 7.98×10^{27} W. An observer on the Earth detects radiation from this star at a distance away of 8.15×10^{16} m.

(i) Calculate the radiant flux intensity F detected by the observer.

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$F = \dots\dots\dots \text{W m}^{-2}$ [2]

- (ii) The mass of hydrogen in the star decreases continuously as a result of the series of nuclear reactions.

Determine the rate of decrease of mass of hydrogen in the star.

rate of decrease of mass = kg s^{-1} [4]

[Total: 11]